ICCV 2011
13th International Conference on Computer Vision
6-13 November 2011, Barcelona, Spain

Exemplar Extraction using Spatio-Temporal Hierarchical Agglomerative Clustering for Face Recognition in Video

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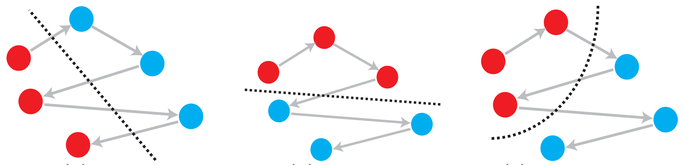
Introduction

- Recent developments in video-based face recognition have resulted in a flurry of approaches that attempt to
  - exploit temporal dynamics by modeling temporal continuity between frames
  - represent multiple independent observations (frames as image sets)
- Unordered set of multiple video frames can be characterized by
  - Subspace-based representation
  - Exemplar-based representation

Contribution of this Work

- Proposal – a new **Spatio-Temporal Hierarchical Agglomerative Clustering (STHAC)** for automatic extraction of face exemplars for face recognition in video sequences
- Two variants:
  - **Global fusion**: unifies spatial and temporal distances between points
  - **Local perturbation**: introduces perturbation of distances in local neighborhood



(a) Spatial (b) Temporal (c) Spatio-temporal

Exemplar-based Setting

- General notation:
  - Training video face sequence of class $\mathbf{c}$:

$$\mathbf{V_c} = \{\mathbf{v_1^c}, \mathbf{v_2^c}, \ldots, \mathbf{v_{N_c}^c}\}, \quad \mathbf{v_i} \in \Re^D$$

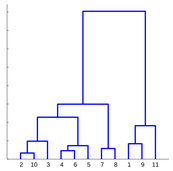
  - Associated exemplar set

$$\mathbf{E_c} = \{\mathbf{e_1^c}, \mathbf{e_2^c}, \ldots, \mathbf{e_{M_c}^c}\}, \quad \mathbf{E_c} \subseteq \mathbf{V_c}, \mathbf{M_c} \ll \mathbf{N_c}$$

  - $\mathbf{N_c}$ frames in video, $\mathbf{M_c}$ extracted exemplars

Obtain Training Exemplars through Clustering

- A necessary step in exemplar-based setting
  - before clustering, apply Locally Linear Embedding (LLE) to uncover manifold structure of video data in lower-dimensional space
  - extract $\mathbf{K}$ number of clusters that group faces of similar appearance
- Hierarchical Agglomerative Clustering (HAC)
  - constructs nested set of partitions represented by cluster tree/dendrogram
  - agglomerative: "bottom up" grouping
  - addresses main limitations of *k-means* clustering – no initial seed selection, no sub-optimal results



STHAC Method

- Defining spatial and temporal distances
  - spatial distance: Euclidean distance, $\mathbf{d_S}$
  - temporal distance: time spanned between two frame occurences:

$$\mathbf{d_T(v_i, v_j)} = |\mathbf{t_{v_i}} - \mathbf{t_{v_j}}|$$

  - compute normalized pairwise distance matrices, $\mathbf{D_S(v_i, v_j)}$ and $\mathbf{D_T(v_i, v_j)}$
- **Global Spatio-Temporal Fusion (GSTF)**
  - blends contribution of spatial and temporal distances using tuning parameter $\mathbf{0 \leq \alpha \leq 1}$, defined by upper and lower bounds $\mathbf{[p_{max}, p_{min}]}$
  - fused spatio-temporal distance defined as:

$$\mathbf{D_{GSTF}} = (\mathbf{p_{max}} - \alpha)\mathbf{D_S} + (\mathbf{p_{min}} + \alpha)\mathbf{D_T}$$

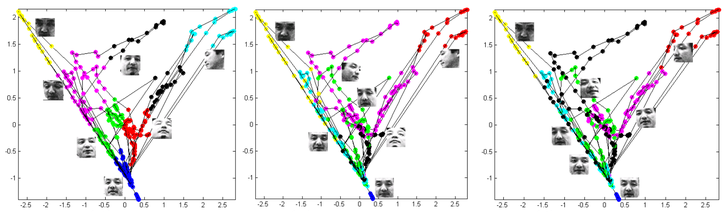
- **Local Spatio-Temporal Perturbation (LSTP)**
  - applies perturbation of spatial and temporal distances based on local neighborhood relationships
  - for each point $\mathbf{v_i}$, we define
    - spatial neighborhood: set of points $\mathbf{Q_{v_i}}$ containing *k*-nearest neighbors
    - temporal neighborhood: temporal window segment set $\mathbf{S_{v_i}} = \{\mathbf{v_{i-w}}, \ldots, \mathbf{v_i}, \ldots, \mathbf{v_{i+w}}\}$ of length $\mathbf{(2w+1)}$
  - common spatio-temporal neighbor: $\mathbf{CSTN_{v_i}} = \mathbf{Q_{v_i}} \cap \mathbf{S_{v_i}}$
  - perturbation affinity matrix:

$$\mathbf{P_{ij}} = \begin{cases} \mathbf{1} - \lambda_{\mathbf{sim}}, & \text{if } \mathbf{x_j} \in \mathbf{CSTN_{x_i}} \\ \mathbf{1} + \lambda_{\mathbf{dis}}, & \text{if } \mathbf{x_j} \in (\mathbf{S_{x_i}} \backslash \mathbf{CSTN_{x_i}}) \\ \mathbf{1}, & \text{otherwise} \end{cases}$$

  - perturbed spatio-temporal distance defined as:

$$\mathbf{D_{LSTP}} = \mathbf{P_{ij}}(\mathbf{D_S} + \mathbf{D_T})$$

- Heuristic selection: number of clusters
  - Ward's criterion for merging clusters
  - Effective – "elbow" of residual error curve



(d) HAC (e) STHAC-GSTF (f) STHAC-LSTP

Face Recognition in Video

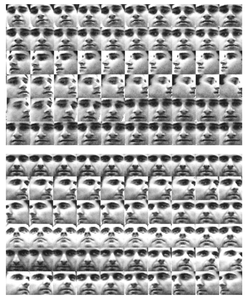
- Feature Representation
  - Neighborhood Discriminative Manifold Projection (NDMP)
- Classification of faces in video
  - Probabilistic Bayes MAP classifier – Decision rule:

$$\hat{\mathbf{c}} = \arg\max_{\mathbf{C}} \; P(\mathbf{c}|\mathbf{V'}) \propto \max_{\mathbf{C}} \; P(\mathbf{c}) \prod_{\mathbf{i=1}}^{\mathbf{N_c}} P(\mathbf{v_i'}|\mathbf{c}) \frac{\sum_{\mathbf{j=1}}^{\mathbf{M}} \mathbf{1/d_{L2}(v_i', e_j^c)}}{\sum_{\mathbf{c=1}}^{\mathbf{C}} \sum_{\mathbf{j=1}}^{\mathbf{M}} \mathbf{1/d_{L2}(v_i', e_j^c)}}$$

where $\mathbf{V'}$ is the test video and $\mathbf{d_{L2}}$ is the *L2*-distance

Experimental Database

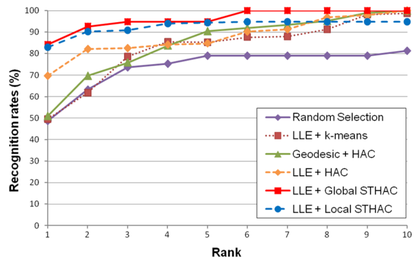
- Honda/UCSD Face Video Dataset
  - first dataset: 59 sequences of 20 people, 300-600 frames/sequence
  - large out-of-plane head rotations, pose and expression variations
  - faces extracted with Viola-Jones cascaded face detector, resized to 32×32 pixels
- Extensive evaluation setup
  - Random sampling of 50 sub-sequences of 200 frames from each test video
  - 1 training video/class, remaining for testing



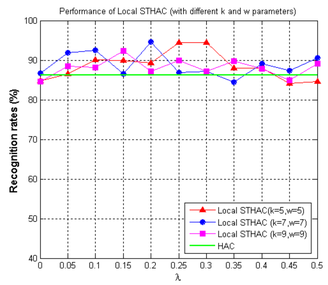
Experimental Results

- Experimental Parameters
  - # of exemplars, $\mathbf{M = 7}$
  - GSTF: $\mathbf{p_{min} = 0.5, p_{max} = 1.5, \alpha = 0.75}$
  - LSTP: $\mathbf{k = 7, w = 7, \lambda = 0.2}$

Method	Average Recognition Rates (%)			
Feature	*PCA*	*LDA*	*NPE*	*NDMP*
Random selection	63.7	64.8	65.7	66.1
LLE + *k*-means	68.5	70.4	65.4	73.7
Geodesic + HAC	73.7	71.3	66.1	76.8
LLE + HAC	66.2	71.2	70.5	86.2
LLE + STHAC-GSTF	74.9	76.9	80.1	**95.0**
LLE + STHAC-LSTP	81.9	87.2	90.8	**94.5**



(g) CMC Curve



(h) STHAC-LSTP vs HAC

Conclusions

- Effective selection of suitable face exemplars (for face recognition in video) using spatio-temporal clustering method (STHAC)
  - Global and local variants were proposed
- Future Work
  - Finding optimum values for free parameters
  - Extend utilization of clusters for image set representations

Due to space constraints, please refer to actual paper for references — Poster created with LaTeX beamerposter

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